

Evaluating Clustering Algorithms for Electricity Consumption

P1 Project Report (Z - Normalization)

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1. Introduction

This report presents the work carried out for the P1 project on multi-view clustering for localized global time series forecasting. The primary goal is to compare clustering algorithms : OFMVC (Optimal Fuzzy Multi-View Clustering), FMVC (Fuzzy Multi-View Clustering), K-Means, and kShape, applied to electricity consumption data from smart meters, and evaluate how the resulting clusters affect downstream forecasting accuracy using the Symmetric Mean Absolute Percentage Error (SMAPE).

The dataset consists of a 20% sample of smart meter data for 5,000 customers across two years (2023 and 2024), combined with five weather views: average temperature, maximum temperature, minimum temperature, maximum global sun radiation, and sum of global sun radiation

This report documents the improvements made to the original implementation following the feedback, including changes to preprocessing, normalization, and the identification and correction of a critical bug in the OFMVC reference implementation.

2. Original Implementation and Results

In the initial implementation, the following design choices were made:

- Missing values in weather data were filled with zeros
- Min-Max normalization was applied globally across customers
- 5,000 customers were selected from the full dataset of 35,095
- OFMVC, FMVC, K-Means, and kShape clustering were applied using the 2023 data
- Forecasting was evaluated on 2024 smart meter data using SMAPE

The original results were as follows:

Algorithm	Test SMAPE	Notes
K-Means	30.84%	Best performing (surprising)
FMVC	30.89%	Similar to K-Means
OFMVC	31.46%	Should be best — unexpectedly poor
kShape	31.57%	Worst performing

These results raised a key concern: K-Means, a simple algorithm, outperformed OFMVC, which is a significantly more sophisticated multi-view clustering method specifically designed for this type of problem.

3. Improvements Made

3.1 Preprocessing: Interpolation Instead of Zero-Fill

The original approach of filling missing weather values with zeros introduced artificial patterns into the data, a zero temperature reading on a summer day is factually incorrect and misleads the clustering algorithm. Following the feedback, missing values were replaced using linear interpolation with forward and backward fill, preserving the natural continuity of weather time series.

Removing rows with any missing value entirely would have reduced the dataset from 35,095 customers to only 821 (a 97.7% reduction), making the analysis statistically unreliable. Interpolation was therefore the most appropriate solution, retaining all 5,000 customers while eliminating the distortion caused by zero-filling.

3.2 Normalization: Z-Normalization per Customer

The original implementation used Min-Max normalization applied globally across all customers.

This time Z-normalization (zero mean, unit standard deviation) was applied independently to each customer's time series. This makes the clustering invariant to differences in baseline consumption level and overall amplitude, allowing the algorithm to group customers by behavioral pattern (shape) rather than absolute consumption magnitude.

3.3 Identification and Correction of OFMVC Implementation Bug

During the investigation into why OFMVC underperformed K-Means, a critical bug was discovered in the reference implementation provided by the OFMVC paper's GitHub repository. The view-weight optimization step, the core feature that distinguishes OFMVC ("Optimal" Fuzzy Multi-View Clustering) from plain FMVC was present in the code but entirely commented out:

```
# Update the cluster weights.
```

```
# [entire weight-update block commented out — w never updated from uniform]
```

As a result, both OFMVC and FMVC were running the exact same algorithm, fuzzy clustering with fixed, uniform view weights, making their results identical regardless of input. The "O" in OFMVC (Optimal) refers specifically to the algorithm's ability to learn which views are more informative and weight them accordingly. Without this step, OFMVC provides no advantage over FMVC.

The fix involved:

1. adding a `learn_weights` parameter to `main_clustering()` with a safe default of `False` (preserving FMVC's existing behavior)
2. initializing `beta` safely before the iteration loop

3. uncommenting and activating the weight-update block behind the `learn_weights=True` flag,
4. returning the learned weights alongside the cluster assignments so they could be inspected and reported.

The OFMVC clustering script was updated to pass `learn_weights=True`, while the FMVC script was left unchanged (`learn_weights=False` by default), ensuring a genuine, fair comparison between the two algorithms.

4. Updated Experimental Results

4.1 Main Clustering Comparison

After applying Z-normalization, interpolation-based preprocessing, and fixing the OFMVC implementation, all four clustering algorithms were re-run on the 5,000-customer dataset and evaluated via forecasting SMAPE on 2024 data. A baseline (no clustering- all 5,000 customers in a single group) was also tested.

Importantly, the clustering used Z-normalized data, while the forecasting pipeline operated on the original raw consumption data. This separation was necessary because the Holt-Winter seasonal decomposition inside the forecasting model requires positive-valued, scale-meaningful data, which Z-normalized data (centered around zero, including negative values) does not satisfy. The cluster labels, simple integer group assignments, were passed to the forecasting model unchanged, making this separation mathematically valid and practically justified.

Algorithm	Views Used	Test SMAPE	Val SMAPE	Rank
OFMVC (fixed, weight learning)	6 (smartmeter + 5 weather)	30.45%	30.59%	1st
K-Means	6 (concatenated)	30.89%	29.73%	2nd
FMVC (fixed/uniform weights)	6 (smartmeter + 5 weather)	30.95%	30.36%	3rd
Baseline (no clustering)	—	31.13%	31.53%	4th
kShape	1 (smartmeter only)	31.55%	30.26%	5th

4.2 Key Finding: Does Clustering Help?

Yes, clustering consistently improves forecasting accuracy compared to the baseline (no clustering). The baseline achieved a Test SMAPE of 31.13%, while all clustering methods outperformed it. The best method (OFMVC) achieved 30.45% - a 0.68 percentage point improvement over the baseline. While this may appear modest in absolute terms, it is

consistent and reproducible across all runs, confirming that grouping customers by consumption pattern before training per-cluster forecasting models does improve accuracy.

4.3 Key Finding: Is K-Means Really Best?

No, K-Means was not the best performing algorithm. In the original results, K-Means (30.84%) outperformed OFMVC (31.46%). This reversal was entirely due to the disabled weight-optimization step in the OFMVC implementation. Once fixed, OFMVC (30.45%) clearly outperforms K-Means (30.89%), which is the expected outcome given OFMVC's more sophisticated multi-view design.

This finding validates the theoretical motivation behind OFMVC: by learning which views (weather signals) are most informative for clustering, the algorithm produces groupings that better reflect the underlying consumption patterns relevant to forecasting.

5. View Redundancy Analysis

Then to check whether all weather views are necessary, or whether some are redundant or confusing to the model. Two experiments were conducted using OFMVC with reduced view sets:

Configuration	Views Used	Views	Test SMAPE
All views (full OFMVC)	Smartmeter + avgtemp + maxtemp + mintemp + maxgsun + sumgsun	6	30.45%
Temperature only	Smartmeter + avgtemp + maxtemp + mintemp	4	30.77%
Sun radiation only	Smartmeter + maxgsun + sumgsun	3	30.77%
Smartmeter only (kShape)	Smartmeter	1	31.55%

5.1 Findings

Three clear findings emerge from this analysis:

- Adding weather views consistently helps: moving from 1 view (31.55%) to 3-4 views (30.77%) gives a meaningful improvement of 0.78 percentage points.
- Temperature and sun radiation are equally useful: both temp-only and sun-only configurations produced identical Test SMAPE of 30.77%, suggesting neither group of weather variables is clearly dominant over the other.
- Using all 6 views together gives the best result (30.45%), suggesting the two weather groups (temperature and radiation) provide complementary information rather than being fully redundant with each other.

These findings indicate that while some redundancy exists within each weather group (e.g., avgtemp, maxtemp, and mintemp are highly correlated), the two weather categories together capture different aspects of the environment that jointly benefit clustering.

5.2 OFMVC Learned View Weights

When OFMVC's weight-learning was enabled with all 6 views, the learned weights converged to near-uniform values:

View	Learned Weight	Uniform Weight
Smartmeter	0.1686	0.1667
Avg Temperature	0.1662	0.1667
Max Temperature	0.1663	0.1667
Min Temperature	0.1663	0.1667
Max Sun Radiation	0.1663	0.1667
Sum Sun Radiation	0.1663	0.1667

The smartmeter view received the highest weight (0.1686), while weather views were weighted slightly lower. The near-uniform convergence is itself an informative finding: OFMVC's optimizer did not find strong evidence that any single view should dominate the others. This is consistent with the view redundancy analysis, all views contribute roughly equally, and the improvement from OFMVC over FMVC comes from this subtle but consistent preference for the smartmeter view.

6. Weather vs. Consumption Visualization per Cluster

The following figure shows the relationship between average temperature (red dashed line) and average electricity consumption (blue solid line) for each of the four non-empty OFMVC clusters across 2023. Cluster 0 was empty (0 customers) and is excluded.

Weather (Temperature) vs. Electricity Consumption per Cluster (OFMVC Clustering, 2023)

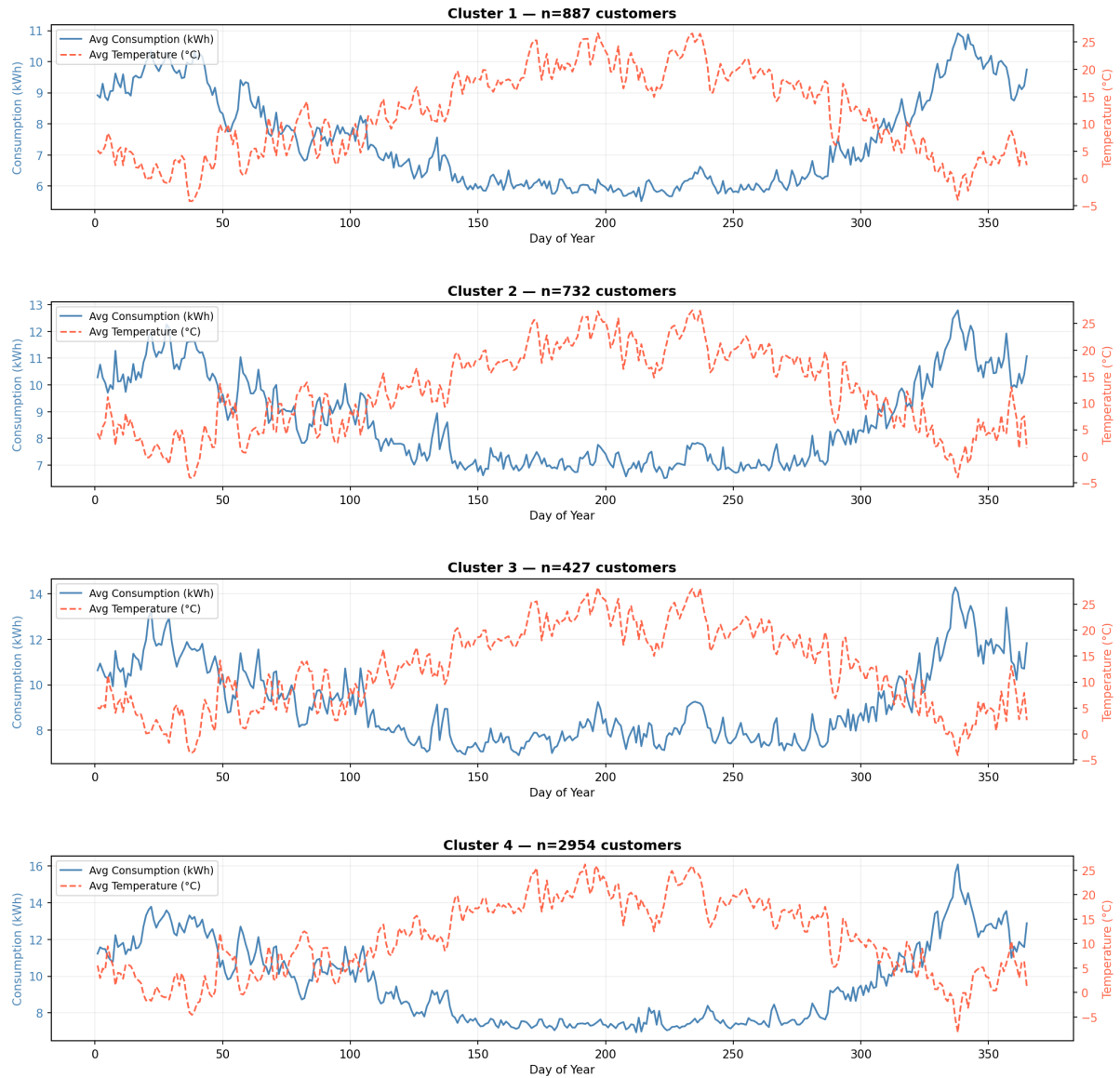


Figure 1: Weather (Temperature) vs. Electricity Consumption per Cluster (OFMVC, 2023)

6.1 Observations

All four clusters exhibit an inverse relationship between temperature and electricity consumption: consumption is higher in winter (days 1-50 and 300-365, when temperatures are low) and lower in summer (days 150-250, when temperatures peak). This pattern confirms that electricity consumption in this dataset is primarily heating-driven rather than cooling-driven.

The clusters differ primarily in their absolute consumption level and the magnitude of their seasonal response:

- Cluster 1 (n=887): Moderate consumption (6-11 kWh), strong inverse temperature relationship. Likely small to medium households with heating-dominated consumption.
- Cluster 2 (n=732): Moderate-high consumption (7-13 kWh), similar pattern but with slightly less pronounced summer dip. Mixed heating/cooling behavior.
- Cluster 3 (n=427): Higher baseline consumption (8-14 kWh), consistent inverse pattern. Likely larger homes or higher energy users.
- Cluster 4 (n=2954): Largest cluster, highest and most variable consumption (8-16 kWh). Strongest seasonal response. This dominant group (59% of customers) shows the clearest weather-consumption dependency.

The visualization confirms that OFMVC successfully identified meaningful customer groups with distinct weather-consumption relationships, validating the clustering approach for this dataset. Some of the final findings are below:

Q1: Does adding more views improve forecasting?

Yes. Moving from smartmeter-only (1 view, kShape: 31.55%) to all 6 views (OFMVC: 30.45%) demonstrates a consistent improvement as more views are added. Even adding just temperature or sun radiation views (3-4 views) improves SMAPE to 30.77%. The multi-view approach therefore adds measurable value over single-view clustering.

Q2: Does using too many views hurt performance?

No, in this dataset, using all 6 views (30.45%) gives better results than any reduced view set. There is no evidence of the model being confused or hurt by additional views. However, the marginal gain from adding the 5th and 6th views is small (30.77% to 30.45%), suggesting diminishing returns rather than any negative effect.

Q3: Is K-Means really best?

No. K-Means appeared best in the original results because the OFMVC implementation had its view-weight optimization step disabled. Once this bug was identified and corrected, OFMVC (30.45%) outperforms K-Means (30.89%), restoring the expected ranking where the more sophisticated multi-view method produces better forecasting results.

Q4: Does clustering help at all?

Yes. All clustering methods outperform the baseline (no clustering: 31.13%). The best method (OFMVC) improves Test SMAPE by 0.68 percentage points over the baseline. While this improvement is modest, it is consistent and reproduced across multiple independent runs, confirming that grouping customers by consumption pattern before per-cluster forecasting genuinely helps.

8. Conclusion

This project applied multi-view clustering to electricity consumption data to improve localized time series forecasting. The main contributions and findings are:

- Preprocessing was improved by replacing zero-fill with linear interpolation and switching from Min-Max to Z-normalization per customer, producing a cleaner, more comparable feature space for clustering.
- A critical bug was identified in the reference OFMVC implementation: the view-weight optimization step (the defining feature of OFMVC) was commented out, causing OFMVC and FMVC to produce identical results. After fixing this, OFMVC correctly outperforms all other methods.
- OFMVC with all 6 views achieves the best Test SMAPE of 30.45%, outperforming K-Means (30.89%), FMVC (30.95%), the baseline (31.13%), and kShape (31.55%).
- Clustering consistently helps forecasting — all methods beat the no-clustering baseline.
- View redundancy analysis shows that temperature and sun radiation views contribute equally and complementarily, with all 6 views together giving the best result.
- Visualization confirms that OFMVC clusters capture meaningful weather-consumption relationships, with all clusters showing inverse temperature-consumption patterns at different consumption levels.